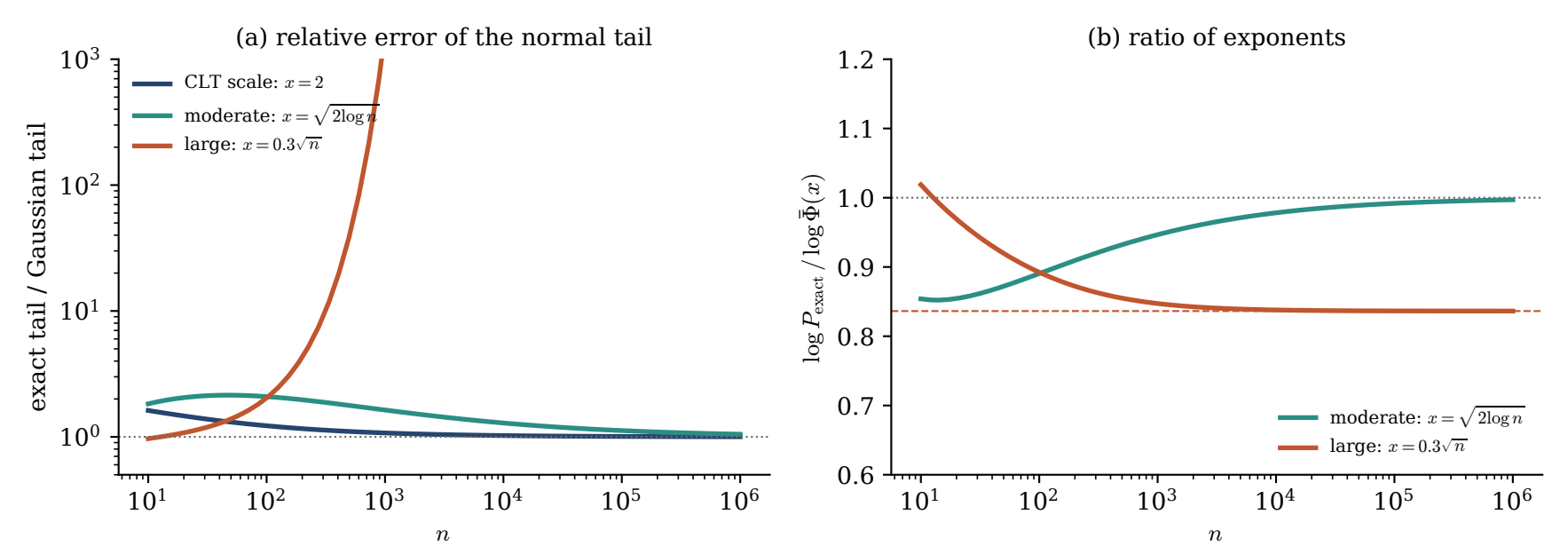
(a) relative error of the normal tail
exact tail / Gaussian tail
10^3
10^2
10^1
10^0
CLT scale: $x = 2$
moderate: $x = \sqrt{2\log n}$
large: $x = 0.3\sqrt{n}$
10^1
10^2
10^3
10^4
10^5
10^6
n
(b) ratio of exponents
$\log P_{\text{exact}} \,/\, \log \bar{\Phi}(x)$
1.2
1.1
1.0
0.9
0.8
0.7
0.6
moderate: $x = \sqrt{2\log n}$
large: $x = 0.3\sqrt{n}$
10^1
10^2
10^3
10^4
10^5
10^6
n